

MODEL QUESTION PAPER

SAFETY IN PROCESS INDUSTRIES

Max.Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. List two corrosive chemicals used in industries.
2. State toxicity.
3. List the personal protective equipment used for electrical works.
4. List the personal protective equipment used for eyes.
5. State chemistry of fire.
6. List the causes of electrical fire.
7. Draw a fire triangle.
8. State the need of work permit system in process plants.
9. State safety checklist.

Part B

II. Answer any EIGHT questions from the following. Each question carries 3 marks

8 X 3 = 24 Marks

1. Classify the chemical process plant hazards with examples.
2. State Explosion and Toxicity Index.
3. List the preliminary hazard analysis methods.
4. Summarize personal protective equipment for head.
5. Select appropriate personal protective equipment in the case of chlorine gas leakage.
6. List the causes of initiation of fire.
7. State the function of a fire alarm.
8. Summarize the functions of fire hydrants.
9. Summarize cold work permit.
10. List the precautions to be taken before entering in to a confined space.

Part C

Answer ALL questions. Each questions carries 7 marks

6 X 7 = 42 Marks

III. Differentiate Threshold Limit Value(TLV) and Permissible Exposure Limit(PEL).

OR

IV. Summarize Flixborough disaster.

V. Enumerate hazard identification techniques used in industries.

OR

VI. Outline fault tree analysis.

VII. Summarize personal protective equipment for whole body.

OR

VIII. Summarize a case study of accident occurred on duty without proper protective devices.

IX. Classify fire according to combustibility of the materials with examples.

OR

X. Enumerate types of fire detectors used in industries.

XI. Illustrate carbon dioxide extinguisher.

OR

XII. Illustrate Dry chemical powder extinguisher.

XIII. Summarize the permit system for the maintenance of equipment.

OR

XIV. Summarize checklist for startup of a chemical plant.

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Course :Safety in Chemical Industries.

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Mark Distribution

Module	Hr/Module	Marks / Module	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	27	2	2	2	6	2	14	6	22
2	18	38	2	2	3	9	4	28	9	39
3	15	33	3	3	3	9	4	28	10	40
4	12	25	2	2	2	6	2	14	6	22
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	27	27
Understanding	93	71
Applying	3	2
Analysing	-	
Evaluating	-	
Creating	-	
Total	123	100

Question Wise Analysis

Course : Safety in Chemical Industries

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.01	R	1	
I.2	M 1.03	R	1	
I.3	M 2.04	R	1	
I.4	M 2.04	R	1	
I.5	M 3.01	R	1	
I.6	M 3.01	R	1	
I.7	M 3.01	R	1	
I.8	M 4.01	R	1	
I.9	M 4.04	R	1	
II.1	M 1.01	R	3	
II.2	M 1.01	R	3	
II.3	M 2.01	R	3	
II.4	M 2.02	U	3	
II.5	M 2.04	A	3	
II.6	M 3.03	R	3	
II.7	M 3.03	R	3	
II.8	M 3.04	U	3	
II.9	M 4.05	U	3	

II.10	M 4.02	R	3	
III.	M 1.02	U	7	
IV.	M 1.04	U	7	
V.	M 2.01	U	7	
VI.	M 2.01	U	7	
VII.	M 2.03	U	7	
VIII.	M 2.04	U	7	
IX.	M 3.01	U	7	
X.	M 3.02	U	7	
XI.	M 3.03	U	7	
XII.	M 3.04	U	7	
XIII.	M 4.01	U	7	
XIV	M 4.04	U	7	